ADAPT-VQE fails with Aer simulator

https://github.com/qiskit-community/qiskit-aqua/issues/1467

ROW 87

ADAPT-VQE fails with Aer simulator #1467

Closed

MariaSapova opened on Dec 2, 2020

Information

- **Qiskit Aqua version:** 0.23.1
- **Python version:** 3.6.8
- **Operating system:** Ubuntu

What is the current behavior?

Adapt-VQE works fine with BasicAer, but fails with Aer backends (both statevector_simulator and qasm_simulator).
In previous qiskit verstion VQEAdapt worked fine.

```
AquaError: 'Circuit execution failed: ERROR: Failed to load qobj: Invalid parameterized qobj: instructio
```

Steps to reproduce the problem

After molecule definition I use built-in methods:

```
backend = Aer.get_backend('qasm_simulator')
quantum_instance = QuantumInstance(backend)
solver = VQEUCCSDFactory(quantum_instance)
adapt = AdaptVQE(transformation, solver)
res = adapt.solve(driver)
```

What is the expected behavior?

Computation runs fine without error, as in BasicAer case

mrossinek on Dec 3, 2020 · Hidden as outdated

Edits

mrossinek added status: can't reproduce status: needs informa... on Dec 3, 2020

mrossinek on Dec 3, 2020 Member

Having found out about the meta package version numbering I know was able to reproduce your error with a stable-branch installation of Qiskit. Let me dig into this a little bit more and get back to you on this.

mrossinek added status: confirmed type: bug and removed status: can't reproduce status: needs informa... on Dec 3, 2020

mrossinek on Dec 3, 2020 Member

PR #1447 fixes this issue and was merged about a week ago. Thus, my initial attempt of reproducing the bug failed, because it is no longer present on the `master` branch.

This means that you have several options:

- wait for the next meta-package release
- install Qiskit Aqua manually at the commit `8583056` (even `0bf9873` is fine, but the commit after that will not work unless you install Terra from the `master` branch, too)

Assignees

No one assigned

Labels

type: bug

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

Ensure aux_operator eigenvalues are normalized
qiskit-community/qiskit-aqua

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Participants

We can also see whether it is possible to release a Qiskit Aqua `0.8.2` based on `0bf9873` which would allow you to install this bug fix via pip but that is not a decision which I can make. @woodsp-ibm @manoelmarques what do you think about this?

mrossinek removed status: confirmed on Dec 3, 2020

MariaSapova on Dec 4, 2020 Author

@mrossinek thank you! I installed master version of qiskit and now it works fine. However, I would like to point out one thing: with qasm simulator aux_operator values grow quadratically with the number of shots. I've already mentioned the same thing for QEOM #1460.

I think that it would be nice to normalize them.

```
In [6]: backend = Aer.get_backend('qasm_simulator')
        quantum_instance = QuantumInstance(backend, shots = 1000, noise_model = None)

        optimizer = SLSQP(maxiter = 200, disp = True)

        solver = VQEUCCSDFactory(quantum_instance, include_custom = True)
        adapt = AdaptVQE(qubitOp_transformation, solver)
        res = adapt.solve(driver)
        print(res)
```

```
=== GROUND STATE ENERGY ===

* Electronic ground state energy (Hartree): -1.857275030145
  - computed part:      -1.857275030145
  - frozen energy part: 0.0
  - particle hole part: 0.0
~ Nuclear repulsion energy (Hartree): 0.719968994449
> Total ground state energy (Hartree): -1.137306035696

=== MEASURED OBSERVABLES ===

  0:  # Particles: 1948676.000 S: 0.000 S^2: 0.000 M: 0.000

=== DIPOLE MOMENTS ===

~ Nuclear dipole moment (a.u.): [0.0  0.0  1.3889487]

  0:
  * Electronic dipole moment (a.u.): [0.0  0.0  1353305.49997601]
    - computed part:      [0.0  0.0  1353305.49997601]
    - frozen energy part: [0.0  0.0  0.0]
    - particle hole part: [0.0  0.0  0.0]
  > Dipole moment (a.u.): [0.0  0.0  -1353304.11102731]  Total: 1353304.11102731
                 (debye): [0.0  0.0  -3439755.62367468]  Total: 3439755.62367468
```

woodsp-ibm on Dec 4, 2020 Member

While you seem to have an appropriate ground state the other operators, such as number of particles (1.948676 million!) and dipole moment are way off (to put it mildly!) @mrossinek Any comments here?

mrossinek mentioned this on Dec 4, 2020

Ensure aux_operator eigenvalues are normalized #1473

mrossinek on Dec 4, 2020 Member

Hi @woodsp-ibm. I was actually just working on a fix together with @Cryoris.
See the new PR #1473

While you seem to have an appropriate ground state the other operators, such as number of particles (1.948676 million!) and dipole moment are way off (to put it mildly!) @mrossinek Any comments here?

mrossinek mentioned this on Dec 4, 2020

Ensure aux_operator eigenvalues are normalized #1473

mrossinek on Dec 4, 2020 Member ···

Hi @woodsp-ibm. I was actually just working on a fix together with @Cryoris.
See the new PR #1473

mrossinek added 2 commits that reference this issue on Dec 4, 2020

`Ensure aux_operator eigenvalues are normalized` … Verified ffcb71a

`Ensure aux_operator eigenvalues are normalized` … Partially verified e72216e

Cryoris linked a pull request that will close this issue Ensure aux_operator eigenvalues are normalized #1473 on Dec 5, 2020

mrossinek added a commit that references this issue on Dec 9, 2020

`Ensure aux_operator eigenvalues are normalized` … Partially verified 49099e2

mrossinek mentioned this on Dec 9, 2020

Ensure aux_operator eigenvalues are normalized Qiskit/qiskit#5496

mrossinek on Dec 10, 2020 Member ···

Since the Aqua core is in the process of being merged into Terra, I re-opened the PR which fixes this over there. See Qiskit/qiskit#5496

mergify added a commit that references this issue on Dec 28, 2020

`Ensure aux_operator eigenvalues are normalized (#5496)` … Partially verified 4d3ed6a

molar-volume added a commit that references this issue on Dec 29, 2020

`Ensure aux_operator eigenvalues are normalized (Qiskit#5496)` … 4e8a076

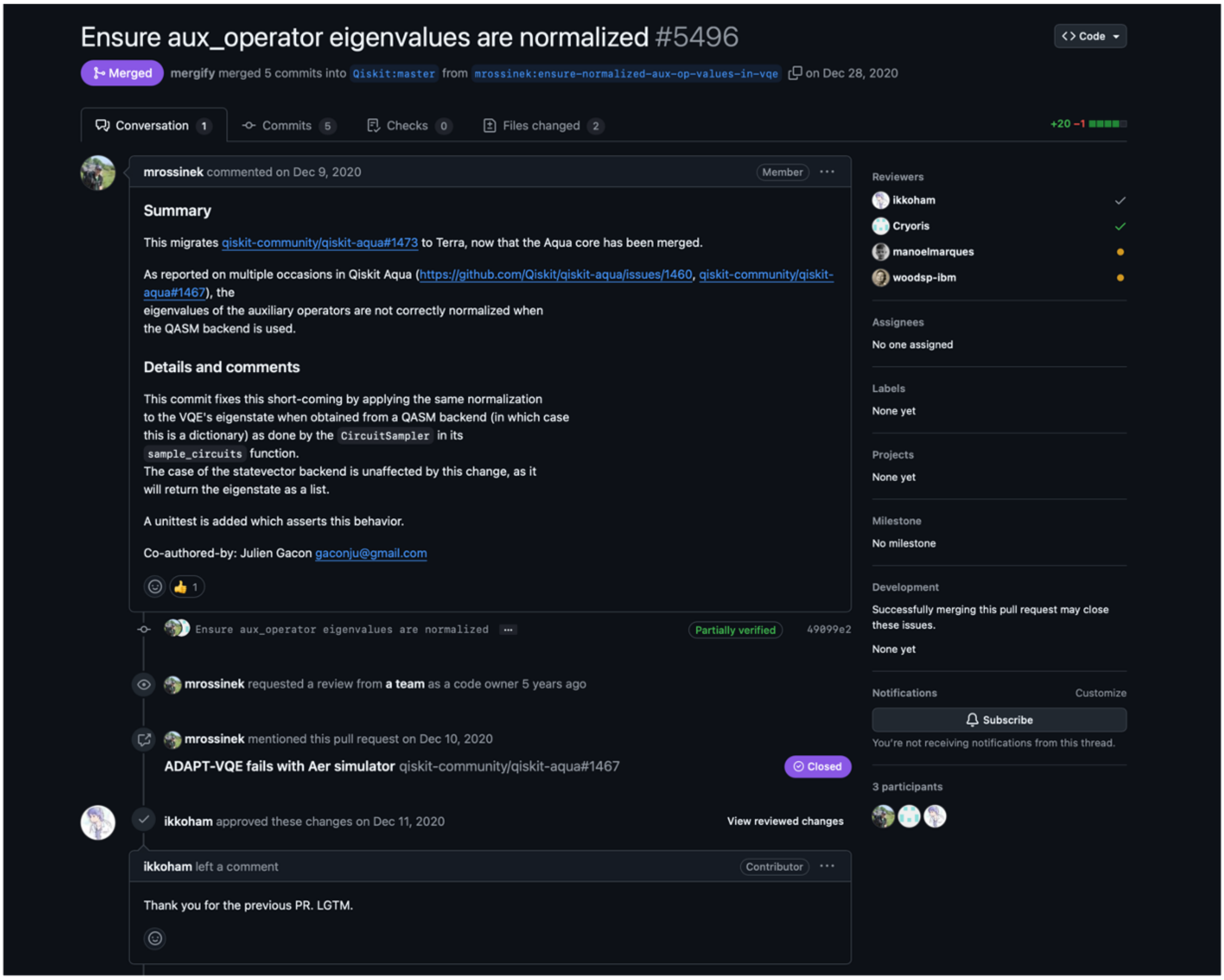
mrossinek on Jan 4, 2021 Member ···

This issue is now also fixed in the migrated Aqua core inside of Terra. Closing.

mrossinek closed this as completed on Jan 4, 2021

mergify added a commit that references this issue on Jan 13, 2021

`Improved handling of ``params`` argument in ``Gradient.convert`` (#55` … Partially verified 799ad14



Ensure aux_operator eigenvalues are normalized #5496

Merged mergify merged 5 commits into Qiskit:master from mrossinek:ensure-normalized-aux-op-values-in-vqe on Dec 28, 2020

Conversation 1 | Commits 5 | Checks 0 | Files changed 2 | +20 −1

mrossinek commented on Dec 9, 2020 — Member

Summary

This migrates qiskit-community/qiskit-aqua#1473 to Terra, now that the Aqua core has been merged.

As reported on multiple occasions in Qiskit Aqua (https://github.com/Qiskit/qiskit-aqua/issues/1460, qiskit-community/qiskit-aqua#1467), the
eigenvalues of the auxiliary operators are not correctly normalized when
the QASM backend is used.

Details and comments

This commit fixes this short-coming by applying the same normalization
to the VQE's eigenstate when obtained from a QASM backend (in which case
this is a dictionary) as done by the `CircuitSampler` in its
`sample_circuits` function.
The case of the statevector backend is unaffected by this change, as it
will return the eigenstate as a list.

A unittest is added which asserts this behavior.

Co-authored-by: Julien Gacon gaconju@gmail.com

👍 1

Ensure aux_operator eigenvalues are normalized … Partially verified 49099e2

mrossinek requested a review from **a team** as a code owner 5 years ago

mrossinek mentioned this pull request on Dec 10, 2020

ADAPT-VQE fails with Aer simulator qiskit-community/qiskit-aqua#1467 — Closed

ikkoham approved these changes on Dec 11, 2020 — View reviewed changes

ikkoham left a comment — Contributor

Thank you for the previous PR. LGTM.

Reviewers: ikkoham, Cryoris, manoelmarques, woodsp-ibm

Assignees: No one assigned

Labels: None yet

Projects: None yet

Milestone: No milestone

Development: Successfully merging this pull request may close these issues. None yet

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3 participants